

Divyansh Shukla

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SUMMARY

Machine Learning Engineer (Systems) with a strong foothold in data pipeline management, model training/fine-tuning, and ML on the edge, with projects spanning flow-matching/diffusion, decoder-only transformers, vision-language model (VLM) experimentation, and robot learning.

EDUCATION

Vellore Institute of Technology

Bachelor of Technology in Computer Science and Engineering (Specialization: AI & ML)

CGPA: 9.03

DAV Public School Sahibabad

XII Standard

91%

DAV Public School Sahibabad

X Standard

95.6%

SKILLS

Languages: Python, C++, CUDA, HTML

Libraries: PyTorch (w/ Distributed), PyTorch Lightning, Hugging Face Transformers, TRL, vLLM, pandas, OpenCV

RL/Sim: Gymnasium, MuJoCo

ML/DL: diffusion models, flow matching, transformer architectures, reinforcement learning, vision-language model experimentation

Tools/Workflow: Hydra, Jupyter notebooks, Modal (remote compute)

PROJECTS

VLM Project | github.com/divyanshuklai/vlm-project

- Built an end-to-end VLM pipeline (data prep → training → eval) that simplified experiment iteration and reduced time-to-result.
- Implemented alignment + preference optimization methods including RLHF (reward modeling + optimization), DPO, and RLVR (GRPO-style RL fine-tuning).
- Enhanced evaluation rigor with benchmark-driven comparisons (competitive with similarly sized models) and standardized, reproducible runs on Modal.

FlappyBirDiT | github.com/divyanshuklai/FlappyBirDiT

- Built a video diffusion pipeline for near real-time inference, conditioned on player inputs to generate responsive gameplay frames.
- Enhanced model selection via ablations over model size, samplers, priors, training methods (SDE/ODE/diffusion-style), and guidance-injection strategies.
- Achieved playable frame quality and frame rate by benchmarking playability targets (throughput + visual fidelity) and iterating toward stable, interactive generation.

Zero-shot ICE RMA (Teacher-Student RL) | github.com/divyanshuklai/zero-shot-ice-rma

- Replicated the original RMA (teacher-student) approach and extended it into a reproducible training + evaluation workflow.
- Increased robustness and generalization via built-in domain randomization and evaluation across a multitude of environments for training and transfer testing.
- Enabled easier sim-to-sim transfer and laid groundwork for sim-to-real deployment tooling (repeatable configs, eval harness, and deployment-oriented utilities).

Fashion Search | github.com/divyanshuklai/fashion_search

- Built a FastAPI conversational chat app for natural-language fashion product discovery across 10 Indian e-commerce stores (Myntra, Ajio, TataCliq, and others).
- Implemented a naive search pipeline (Tavily web search → Gemini 2.5 Flash parsing → BeautifulSoup scraping) to turn search snippets into structured product data.
- Engineered a priority-based image-extraction strategy (JSON-LD, meta tags, keyword-matched img tags, OG fallback) to reliably surface product images and prices in a chat UI; extending to an agentic search variant with planning, memory, and tool use for comparison against the naive baseline.

RELEVANT KNOWLEDGE

Applied Accelerated Artificial Intelligence (NPTEL25CS98)

Deep Learning with Computer Vision (NPTEL25CS93)

Deep Learning Specialization (Coursera)